KPS PUBLIC SCHOOLS • STUDENT TECH BENCH PROGRAM

# CTL Chromebook PX14E / EX / EXT

## Solder-Free Repair Guide



Every field-replaceable repair that needs a screwdriver, not a soldering iron.
CTL Chromebook PX14E, PX14EX, PX14EXT (touch), including the Enterprise versions.

### In this guide


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Built from CTL's parts and spec sheets and standard 14-inch construction.
Part numbers: confirm with CTL support by serial before ordering.

# 1 Read first

This guide covers every CTL Chromebook Enterprise/PX14E/PX14EX/PX14EXT repair the Tech Bench can do with hand tools: swapping a part, reseating a cable, or fixing software. Anything that needs a soldering iron, hot air, or board-level work is in Section 6 and goes to the lead tech or the manufacturer.

**How this guide was built.** CTL publishes spec sheets and sells spare parts for this model, but no public step-by-step service manual. This guide follows the standard bottom-access 14-inch design used by CTL's other Chromebooks. Screw counts say "count them"; record them in the first-teardown notes boxes and check against CTL support if unsure.

**No public service manual.** The maker hasn't published a service manual for this model. The steps follow its documented sister design and standard Chromebook construction. Where a step says "count them", count on your first teardown and write it in the notes box. If something doesn't match, stop and ask the lead tech.

**Variants.** The letters after the model (E / EX / EXT) change the processor or add touch. Write your exact model on the first-teardown boxes; the display part differs for touch models.

## Bench levels

| Level | Who may do it | Typical repairs |
|---|---|---|
| LEVEL 1 | Any trained student | Software fixes, single keys, bottom cover |
| LEVEL 2 | Student with supervisor sign-off on this repair | Battery, Wi-Fi, touchpad, speakers, I/O board, display parts, hinges |
| LEVEL 3 | Lead tech or adult supervisor only | Heat sink, upper case (keyboard) transplant, liquid damage, system board |

## Before you touch any device

1. ☐ Check the asset tag and look the device up in the district inventory. Note any open ticket.
2. ☐ Ask the lead tech whether the unit is still under CTL warranty or an accidental-damage plan. A warranty claim may cost less than a part, and opening a unit can affect warranty coverage.
3. ☐ Photograph all four sides of the device before opening it.
4. ☐ Record the symptom in the user's own words on the ticket.
5. ☐ Warn the user that a reset erases local files (Downloads). Cloud files in Drive are safe.

## Safety rules

- **Power off, unplug, remove any SD card and cables,** then disconnect the battery before any other work inside.
- **Lithium batteries:** never puncture, bend, or pry against the battery. A swollen, hot, hissing, or smoking battery is a Level 3 stop. Move it to the metal battery bin or sand and get an adult.
- **ESD:** wear a grounded wrist strap on an ESD mat. Touch boards only by the edges.
- **Ribbon cables:** unlock every ZIF latch before sliding a ribbon out. A locked ribbon tears.
- **Glass:** wear safety glasses when removing a cracked screen.
- **Before power-on:** check for loose screws inside. Shake gently and listen. A loose screw can short the board.

## Tools and supplies

| Item | Use | Notes |
|---|---|---|
| Phillips #0 and #1 | All screws | Magnetic tip helps |
| Plastic spudger, opening picks, plastic tweezers | Clips, latches, ribbons | Only plastic touches connectors |
| ESD mat + wrist strap | Static protection | |
| Magnetic screw mat | Keeping screws in order | Screws are often different lengths |
| 90%+ isopropyl alcohol + soft brush | Cleaning contacts, spill residue | |
| Kapton tape | Re-taping cables and connectors | |
| Recovery USB or SD card, 8 GB+ | ChromeOS recovery | Chromebook Recovery Utility |
| Known-good USB-C charger | Testing power and charging | |
| Safety glasses | Cracked-screen removal | |
| Fire-safe battery bin or sand bucket | Damaged batteries | Required at every bench |

# 2 Know the machine

## Specs that matter for repair

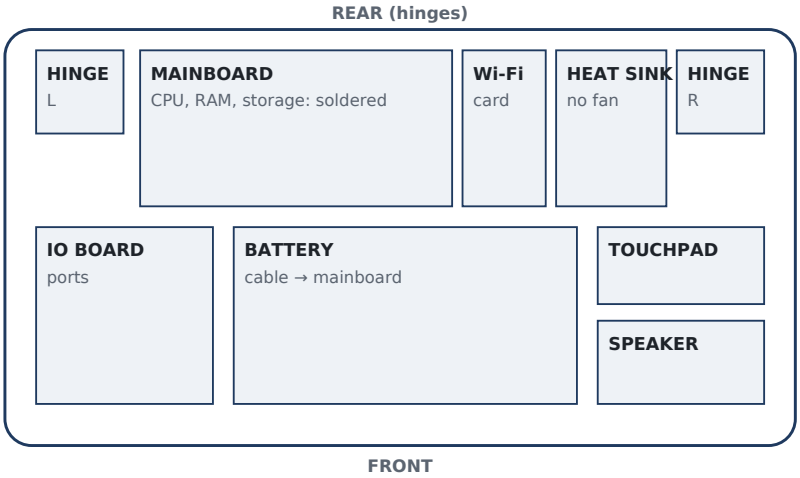
| Item | PX14E | Why it matters |
|---|---|---|
| Processor | Intel Celeron / Pentium (N-series), by variant | Soldered |
| Memory / storage | Soldered | System board level |
| Display | 14" FHD; EXT = touch | Match the panel |
| Access | Bottom cover, Phillips screws + clips | Check under the rubber feet |
| Wireless | Check: socketed card or on the board | See the Wi-Fi repair |
| Cooling | Passive (no fan) on most variants | |

## How it comes apart

A screwed-on bottom cover comes off first. Under it: battery, system board, heat sink, I/O board, speakers and touchpad. The keyboard is built into the upper case (palm rest), the last part left after everything else is removed.

## Inside the base

With the cover off you see the parts below. It's a rough map; correct it on your first teardown.




*ILLUSTRATIVE — block positions are approximate. Confirm on your first teardown.*

## PX14E parts list

CTL part numbers vary by configuration. Look up the exact part by serial number before ordering.

| Part | Part number | Repair |
|---|---|---|

**Look up by serial.** CTL part numbers vary by configuration. Confirm with CTL support or your district's parts vendor using the serial number.

# 3 Triage: symptom → repair

Start with the cheapest, least invasive fix. Work through "try first" before ordering parts.

| Symptom | Try first | Then | Part likely needed |
|---|---|---|---|
| Won't turn on, no lights | Known-good charger 30 min → S1 | R4 → R3 | Battery, or system board |
| Not charging | Other USB-C port and charger → S1 | R3 → Sec. 6 if a port is dead | Battery or system board |
| Battery drains fast / low health in S2 | S2 battery test | R3 | Battery |
| Swollen battery / bulging case | STOP — Level 3 | R3 by lead tech | Battery |
| Black screen, power light on | External display via USB-C; S1 | R4 → R13 → R12 | Display cable or panel |
| Flicker or lines that change with lid angle | S1 | R13 | Display cable |
| Cracked or blotchy screen | — | R12 | LCD panel |
| Bezel loose or broken | Reseat clips | R11 | Bezel |
| Lid floppy, hinge cracked | Check hinge screws | R15 | Hinge(s) |
| One key missing or broken | — | R1 | Keycap |
| Several keys or a row dead | S2 keyboard test; S1; S4 | R4 → R16 | Keyboard / top cover |
| Keys type wrong symbols | S4 | S4 | None |
| Touchpad dead or erratic | S1; check touchpad settings | R4 → R6 | Touchpad or its cable |
| USB-A / SD card / audio dead | Try another device in the port; S1 | R8 | I/O board or its cable |
| Camera not detected | Privacy shutter open? S1 | R14 | Camera module |
| No sound (headphones work) | Volume, output; S1; S3 | R7 | Speakers |
| Wi-Fi missing or weak | S1 → S3 | R5 | Wi-Fi card or antennas |
| Runs hot, fan loud, shuts down | Check vents for lint; S2 | R9 | Fan or heat sink |
| Liquid spill | Power off NOW, unplug | R17 | Depends on damage |
| "ChromeOS is missing or damaged" | S3 recovery | R18 | System board (storage) |
| Cracked bottom or lid shell | — | R2 | Cover |

# 4 Software repairs

No tools needed. Always do S1 before opening a device.

## S1 Hard reset (EC reset) — LEVEL 1

| | |
|---|---|
| **Fixes** | Won't power on, won't charge, frozen, black screen, unresponsive keyboard or touchpad |
| **Time** | 2 minutes |
| **Tools** | Known-good USB-C charger |
| **Data loss** | None |

A hard reset restarts the embedded controller (power, keyboard and charging chip). It erases nothing.

### Procedure

1. ☐ Unplug the charger and any USB devices. Remove any SD card.
2. ☐ Hold Power for 10 seconds so the device is fully off.
3. ☐ Hold **Refresh** (circular arrow), then tap **Power**. Release Refresh when it starts.
4. ☐ If nothing happens: plug in a known-good charger, wait 30 minutes, and repeat with the charger connected. Try the other USB-C port too.

### Verify before closing the ticket

- Device reaches the login screen.
- Charging light comes on with the charger connected.

**Tip.** After a battery has been disconnected, plug in a charger before deciding the device is dead. Many Chromebooks won't wake on the power button alone.

## S2 Built-in hardware diagnostics — LEVEL 1

| | |
|---|---|
| **Fixes** | Confirms the fault before ordering a part: battery, CPU, memory, keyboard, touchpad, Wi-Fi |
| **Time** | 5–15 minutes |
| **Tools** | None |
| **Data loss** | None |

### Procedure

1. ☐ Sign in with a bench test account if you can.
2. ☐ Open the Launcher, search **Diagnostics**, and open it.
3. ☐ **System:** read battery health and cycle count, then run the battery tests.
4. ☐ Run the CPU and memory tests.
5. ☐ **Connectivity:** confirm Wi-Fi is detected and run the tests.
6. ☐ **Input:** open the keyboard tester and press every key; test the touchpad and touchscreen.
7. ☐ Screenshot any failed result (Ctrl + Show windows) and attach it to the ticket.

### Verify before closing the ticket

- Results recorded on the ticket.
- Parts ordered only for a failed test or a fault you saw yourself.

## S3 Powerwash and USB recovery — LEVEL 1

| | |
|---|---|
| **Fixes** | Slow, crashing, corrupted OS, "ChromeOS is missing or damaged", Wi-Fi missing after an update |
| **Time** | Powerwash 10 min; recovery 20–40 min |
| **Tools** | Recovery USB or SD card (8 GB+) |
| **Data loss** | **Yes.** Erases all local files (Downloads). |

**District-managed devices.** Enrolled devices should force re-enrollment on their first Wi-Fi connection after a wipe. Confirm with the lead tech that forced re-enrollment is on and the bench Wi-Fi reaches enrollment. Never enter Developer Mode on a district device.

### Procedure

1. ☐ **Check for updates first:** Settings > About ChromeOS > Check for updates.
2. ☐ **Powerwash:** sign out, press Ctrl + Alt + Shift + R, choose Restart, then Powerwash. (May be blocked by policy; use recovery instead.)
3. ☐ **Make recovery media:** on a working computer, open the Chromebook Recovery Utility, choose the **CTL Chromebook PX14E** from its list (or search the model), insert an 8 GB+ drive, and click Create now.
4. ☐ Hold **Esc + Refresh**, tap **Power**, and release when the recovery screen appears.
5. ☐ Insert the recovery media and follow the prompts.
6. ☐ After it reboots, connect to Wi-Fi and confirm it re-enrolls to the KPS domain.

### Verify before closing the ticket

- Boots to the enrollment or KPS sign-in screen.
- Shows as enrolled in the Admin console.
- Recovery can't write to storage: storage has failed. That's a system board repair or warranty claim.

## S4 Keyboard layout and international keyboard fix — LEVEL 1

| | |
|---|---|
| **Fixes** | Keys type the wrong symbols (@ and " swapped, accents appear, Z and Y swapped); a replacement keyboard with a different layout |
| **Time** | 5 minutes |
| **Tools** | None |
| **Data loss** | None |

ChromeOS can have several keyboard layouts installed. Students often switch layouts by accident with **Ctrl + Shift + Space**, or have **US International** on, which turns ' and " into accent keys. The physical keyboard is fine.

### Procedure

1. ☐ Look at the bottom-right shelf. A layout code (US, INTL, UK, ES...) means more than one layout is installed.
2. ☐ Open **Settings > Device > Keyboard and inputs > Input** (on older versions: Settings > Advanced > Languages and inputs > Inputs).
3. ☐ Set **English (US)** as the active input. Remove any layouts the student doesn't need, especially US International, unless a teacher asked for it.
4. ☐ Test in a text box: Shift + 2 should type @ and Shift + ' should type ".
5. ☐ If a **replacement keyboard** has a different physical layout (UK keyboard on a US unit, for example), order the correct US part. Changing the software layout won't fix printed key labels.
6. ☐ If the wrong layout keeps coming back for everyone who signs in, it's set by district policy. Tell the lead tech; don't change it on the device.

### Verify before closing the ticket

- Every key types its printed character in the S2 keyboard tester and in a text box.
- Only the needed layouts remain installed.

# 5 Hardware repairs

Most repairs start with the bottom cover (R2) and disconnecting the battery (R3).

**Golden rules.** Photograph every layer before removing it. Lay screws on the mat in the same pattern as the device. Unlock every connector before pulling a cable. Never force a part: if it won't move, find the screw or clip you missed. Test before the last screws go in.

## R1 Single keycap replacement — LEVEL 1

| | |
|---|---|
| **Fixes** | Missing, cracked or loose key |
| **Part** | Individual key (ask the manufacturer or harvest from a scrapped keyboard) |
| **Time** | 5 minutes |
| **Tools** | Plastic pick, tweezers |

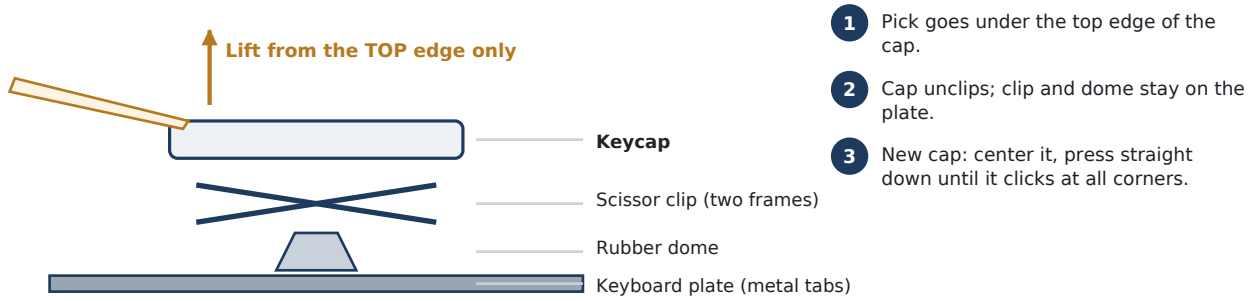



*Illustration: side view of one key. Not to scale.*

### Procedure

1. ☐ Power off. Open the lid flat on the mat.
2. ☐ Slide a plastic pick under the TOP edge of the damaged key and lift until it unclips.
3. ☐ Check the scissor clip and rubber dome under it. A broken clip must be replaced too.
4. ☐ Fit a clip: hook its pins under the metal tabs on the keyboard plate, then press flat.
5. ☐ Center the new key over the clip and press straight down until it clicks at every corner.
6. ☐ Wide keys (Space, Shift, Enter): seat the metal stabilizer wire in its holders first.

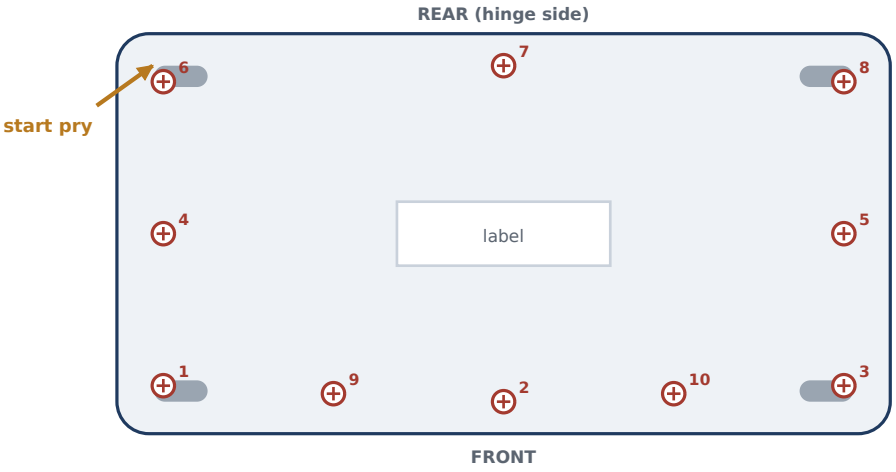
### Verify before closing the ticket

- Key clicks evenly and springs back.
- Key registers in the S2 keyboard tester.

# R2 Base cover (bottom cover) removal

LEVEL 1

| | |
|---|---|
| **Fixes** | Cracked or missing bottom cover; the first step for most internal repairs |
| **Part** | Bottom cover (look up by serial) |
| **Time** | 10 minutes |
| **Tools** | Phillips driver, plastic picks, screw mat |




1. Count the bottom screws.
2. Lay screws on the mat in this same pattern.
3. Pry from the corner shown, then work around the edge.
4. Disconnect the battery before anything else.

*Illustration: underside. Positions illustrative; count on first teardown.*

Check for screws under the rubber feet before prying.

## Procedure

1. ☐ Shut down fully, unplug, and remove any SD card.
2. ☐ Close the lid and turn the device upside down on the mat.
3. ☐ Remove the bottom cover screws (count them; record the number in the notes box).
4. ☐ Release the cover with a plastic pick, starting at a rear corner, then work around the sides. Lift it off.
5. ☐ **Disconnect the battery now** (battery repair, first steps).
6. ☐ To refit: line up the cover, press around the edges until every latch snaps, then refit every screw. Snug only.

## Verify before closing the ticket

- No gaps around the seam; device sits flat.
- Every screw refitted, none left over.

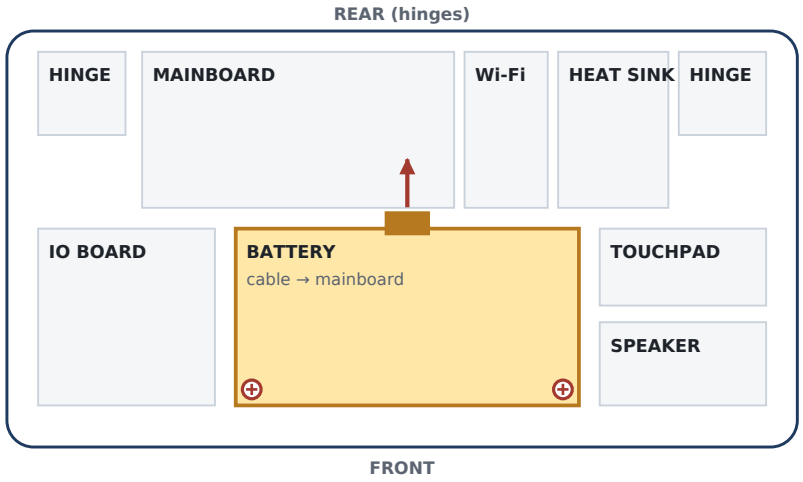
**First-teardown notes** — fill in the first time you do this repair

Number of screws, and which holes take the longer ones? _____

# R3 Battery replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Won't hold charge, low health in S2, swollen battery |
| **Part** | Battery (look up by serial) |
| **Time** | 15–20 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, ESD strap |




1. Unplug the battery cable from the mainboard first (red arrow).
2. Pry at the plug body, never the wires.
3. Remove the battery screws (count them).
4. Lift out flat. Never bend or pry against the cell.

*Illustration: base interior, battery highlighted. Screw positions illustrative.*

**Swollen battery = Level 3.** Don't press or pry. The lead tech handles it; the old cell goes straight to the fire-safe bin.

## Procedure

1. ☐ Remove the bottom cover.
2. ☐ Lift any tape over the battery connector and disconnect the battery cable from the system board.
3. ☐ Remove the battery screws (count them; record the number in the notes box).
4. ☐ Lift the battery out. Never bend it.
5. ☐ Fit the new battery, refit the screws, reconnect the cable.
6. ☐ Refit the cover. Plug in the charger before powering on.

## Verify before closing the ticket

- Charging light comes on; battery percentage shows.
- S2 battery health near 100% on a new battery.
- Old battery goes to battery recycling, never the trash.

# R4 Reseating internal connectors

LEVEL 2

| | |
|---|---|
| **Fixes** | Dead keyboard, touchpad or speakers; black screen; dead ports; no power after a drop |
| **Part** | None, unless a cable is damaged |
| **Time** | 15–25 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol |

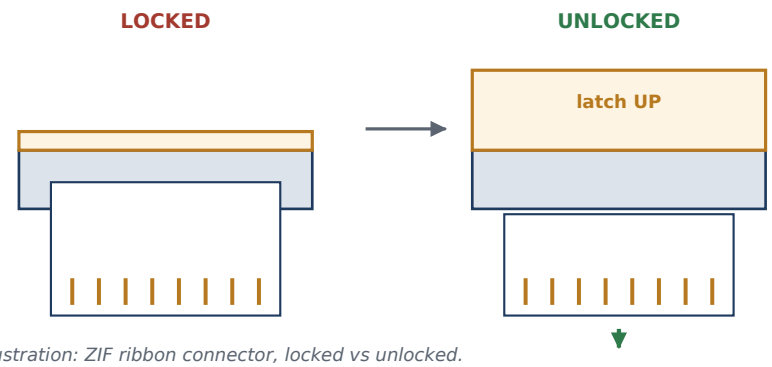



*Illustration: ZIF ribbon connector, locked vs unlocked.*

1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

| Connector | Type | How to release |
|---|---|---|
| Battery cable | Board-to-wire | Lift tape; pry at the connector body |
| Keyboard ribbon | ZIF ribbon | Unlock the latch; slide out |
| Touchpad ribbon | ZIF ribbon | Unlock the latch; slide out |
| Display (eDP) cable | Locking connector | Lift tape and the locking bar, pull straight |
| I/O board cable(s) | ZIF / FFC | Unlock both ends first |
| Speaker cable | Board-to-wire | Pry at the plug body |
| Wi-Fi antennas (2) | Snap-on coax | Lift straight up; note Main/Aux |

## Procedure

1. ☐ Open the device and disconnect the battery.
2. ☐ Hold Power for 5 seconds with the battery disconnected to drain leftover power.
3. ☐ For each connector related to the symptom: photograph it, unlock and release it, and inspect it for bent pins, burn marks or creases.
4. ☐ Clean dirty contacts with a brush dampened in isopropyl alcohol. Let them dry 5 minutes.
5. ☐ Reconnect each one. Ribbons must go in straight and all the way before you close the latch.
6. ☐ Reconnect the battery last. Plug in and test **before** closing up.

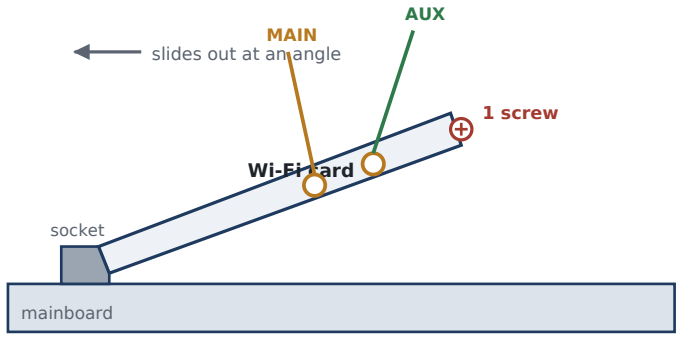
## Verify before closing the ticket

- Symptom gone in a full test, not just at boot.
- Creased or torn ribbons replaced.

# R5 Wi-Fi card reseat or replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | No Wi-Fi or Bluetooth, weak signal, Wi-Fi gone after a drop |
| **Part** | Wi-Fi card (look up by serial) |
| **Time** | 15 minutes |
| **Tools** | Phillips driver, plastic tweezers |




1. Photograph which antenna goes on which terminal.
2. Lift both antenna plugs straight up with plastic tweezers.
3. Remove the 1 screw. The card springs up at an angle.
4. Slide it out of the socket. Refit in reverse.

*Illustration: M.2 Wi-Fi card, side view. Colors of antenna leads vary.*

Check first: on some builds the Wi-Fi is soldered to the system board. If there's no socketed card under a bracket, Wi-Fi is a system-board repair (Section 6).

## Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Find the Wi-Fi card.
3. ☐ Note which antenna goes where (photograph them), then lift both straight off.
4. ☐ Remove the Wi-Fi card screw (count them; record the number in the notes box).
5. ☐ The card tilts up. Slide it out of the socket at an angle.
6. ☐ Refit at an angle, press down, refit the screw, snap on both antennas and test.

## Verify before closing the ticket

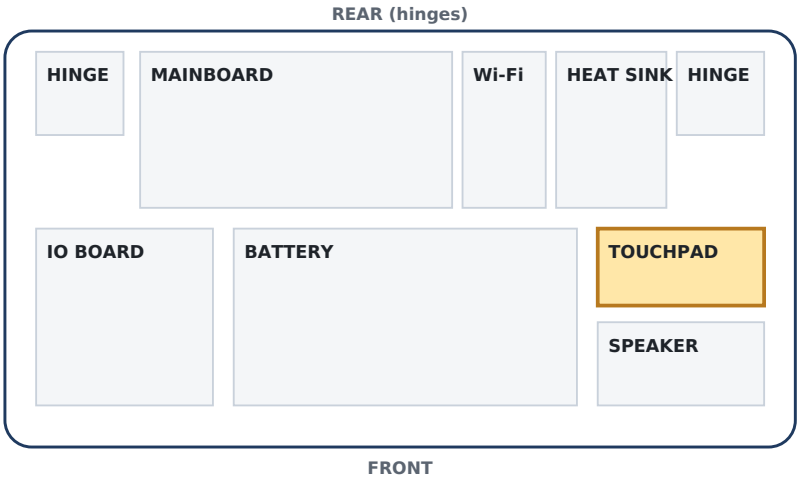
- Wi-Fi connects and the S2 connectivity tests pass.
- Bluetooth finds a device.
- Weak signal after a lid repair usually means a pinched antenna lead.

**Approved cards only.** Use the CTL spare part for this model. Some firmware rejects other cards.

# R6 Touchpad and touchpad cable replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Touchpad dead, erratic, cracked or won't click (keyboard works) |
| **Part** | Touchpad (look up by serial) |
| **Time** | 20–30 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, Kapton tape |




1. Battery disconnected first.
2. Unlock the ZIF latch before sliding the ribbon out.
3. Lift tape or Mylar; remove the screws.
4. Conductive tape must touch the pad on reassembly.

*Illustration: base interior, touchpad highlighted. Position illustrative.*

Check first: touchpad turned on in Settings, S1 done, and the cable reseated.

## Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Unlock the ZIF latch and slide the touchpad ribbon out.
3. ☐ Lift any Mylar or tape, then remove the touchpad bracket screws (count them; record the number in the notes box).
4. ☐ Lift the touchpad out.
5. ☐ Fit the new touchpad, refit the screws and tape, reconnect the ribbon and lock the latch.

## Verify before closing the ticket

- Cursor moves, clicks and two-finger scrolls.
- Click works at all four corners.
- Passes the S2 touchpad test.

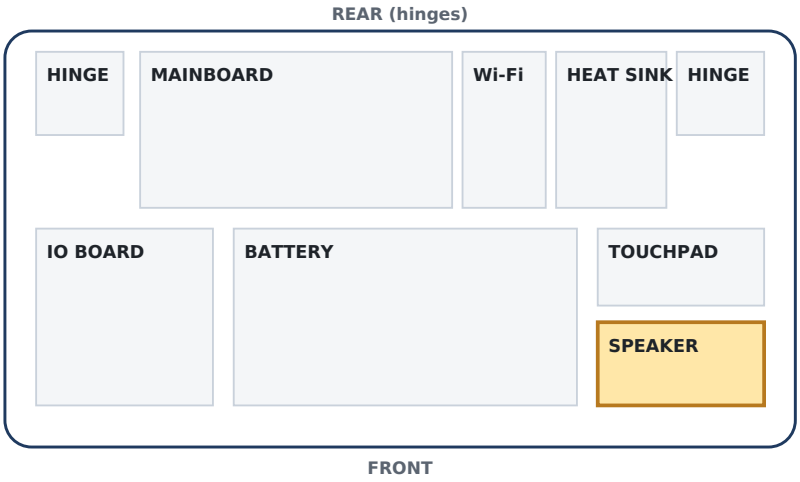
**First-teardown notes** — fill in the first time you do this repair

Touchpad screw count and any tape positions. _____

# R7 Speaker replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | No sound or crackling from the speakers, with headphones working |
| **Part** | Speakers, left and right pair (look up by serial) |
| **Time** | 20 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers |




1. Battery disconnected.
2. Unplug the speaker cable at the board.
3. Unroute the cable, noting each clip.
4. Keep the rubber isolators: they stop rattle.

*Illustration: base interior, speaker highlighted. Position illustrative.*

Check first: if headphones also have no sound, the fault is software or the system board, not the speakers.

## Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Disconnect the speaker cable from the system board and free it from its routing clips.
3. ☐ Remove the speaker screws (count them; record the number in the notes box).
4. ☐ Lift both speakers out together, keeping the rubber isolators.
5. ☐ Fit the new pair, route the cable exactly as before, reconnect.

## Verify before closing the ticket

- Clear sound from both sides with no rattle at full volume.
- Headphone jack still works.

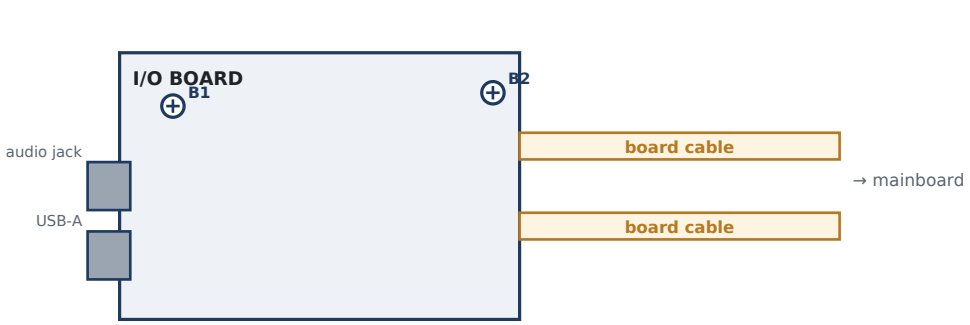
**First-teardown notes** — fill in the first time you do this repair

Speaker screw count. _____

# R8 I/O (USB) daughterboard replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Dead USB-A, audio jack with the system board working |
| **Part** | I/O (USB) board and cable (look up by serial) |
| **Time** | 20–30 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers |




1. Remove the board screws (B).
2. Unlock both connectors, then slide out the 30- and 40-pin ribbons.
3. Lift the board, tilting ports out of the case.

*Illustration: USB board from above. Port order and screw spots illustrative. Confirm on yours.*

## Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Unlock and disconnect the I/O board ribbon(s) at both ends.
3. ☐ Remove the I/O board and bracket screws (count them; record the number in the notes box).
4. ☐ Lift the board out, fit the new one, refit the bracket and screws, reconnect the ribbons.

## Verify before closing the ticket

- Every port on the board works (flash drive, charger, SD card, headphones as fitted).
- Bracket and all screws refitted.

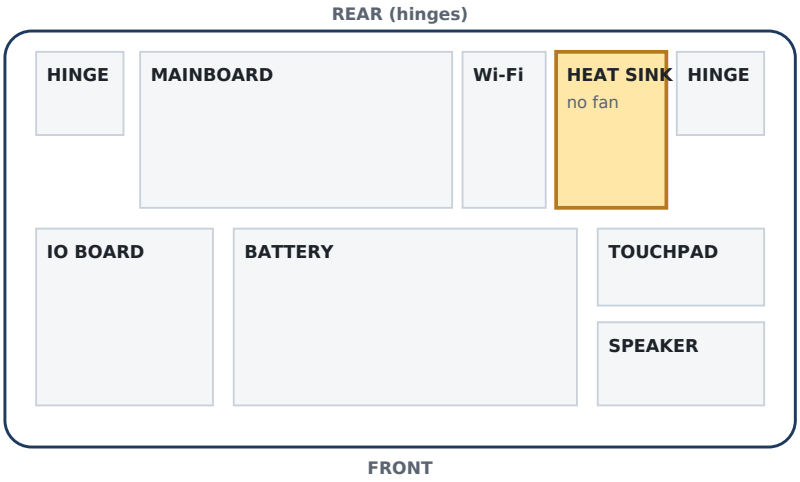
**First-teardown notes** — fill in the first time you do this repair

I/O board screw count and what ports it carries. _____

# R9 Heat sink replacement

LEVEL 3

| | |
|---|---|
| **Fixes** | Overheating, thermal shutdowns, loud or rattling fan |
| **Part** | Heat sink (look up by serial) |
| **Time** | 20–30 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol, lint-free cloth |




1. Battery disconnected.
2. Loosen the screws in the numbered order printed on the heat sink.
3. Clean old thermal paste with alcohol.
4. New thermal material, then tighten in order.

*Illustration: base interior, cooling highlighted. Position illustrative.*

**Thermal material.** Clean old thermal material off the processor and heat sink every time, and fit new material. Never run the board without the heat sink.

## Procedure

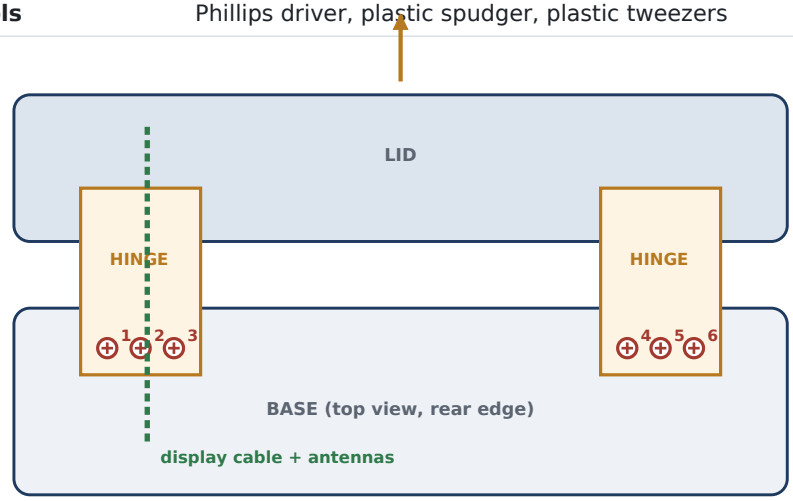
1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Loosen the heat sink screws (count them; record the number in the notes box) in the order printed on the heat sink.
3. ☐ Lift off the heat sink. Clean all old thermal material off the processor and heat sink.
4. ☐ Apply new thermal material, then refit, tightening in order.

## Verify before closing the ticket

- Device runs a 10-minute video without shutting down.
- Fan (if fitted) spins up under load and is quiet at idle.

## R10 Display assembly removal (separating lid from base) LEVEL 2

| | |
|---|---|
| **Fixes** | Needed for hinge, lid and some board repairs; replacing a whole display assembly |
| **Part** | Display assembly or individual display parts (look up by serial) |
| **Time** | 20–30 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers |




1. Battery out, display cable unplugged at the mainboard.
2. Unthread cable and Wi-Fi antennas from the guides FIRST.
3. Remove the display-to-base hinge screws (count them).
4. Support the lid, then lift it away from the base.

*Illustration: hinge area seen from above with the D cover off. Positions approximate.*

### Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Disconnect the display cable (lift the tape and locking bar) and both Wi-Fi antennas; free them from their routing.
3. ☐ Open the lid to 90° or more and support it. Remove the hinge screws in the base (count them; record the number in the notes box).
4. ☐ Lift the display assembly away from the base.
5. ☐ Refit in reverse; route the antennas and display cable back through every clip.

### Verify before closing the ticket

- Display, camera and Wi-Fi all work after reassembly.
- Lid opens through its full range without pinching cables.

# R11 Display bezel replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Cracked, loose or missing screen bezel; the first step for panel, camera and display-cable repairs |
| **Part** | Display bezel (look up by serial) |
| **Time** | 10 minutes |
| **Tools** | Plastic picks, Phillips driver |

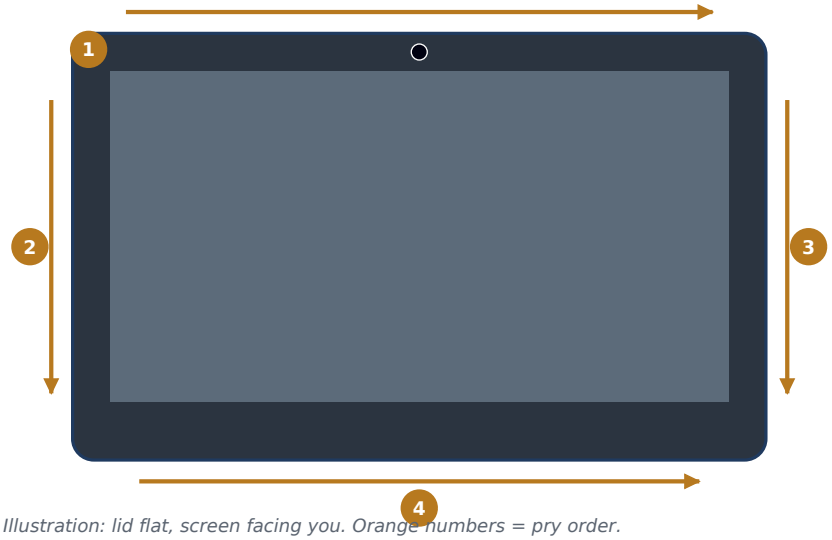



*Illustration: lid flat, screen facing you. Orange numbers = pry order.*

1. No screws: this bezel clips on.
2. Start at the top edge with a plastic pick.
3. Work down the left and right sides.
4. Do the bottom (hinge) edge last.
5. Don't pull on the camera window.

## Procedure

1. ☐ Open the lid flat.
2. ☐ Starting at the bottom inside edge, work a plastic pick around the bezel to release the clips.
3. ☐ Lift the bezel off. Don't pull on the camera window.
4. ☐ Refit by pressing it in until every clip seats.

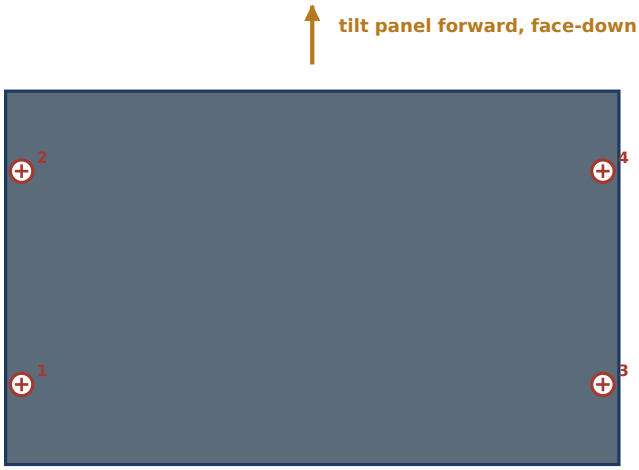
## Verify before closing the ticket

- Bezel sits flush all round.
- Camera window lined up; privacy shutter slides freely.
- Lid closes fully.

# R12 LCD panel replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Cracked screen, dead pixels, blotches, black screen not fixed by the display cable |
| **Part** | LCD panel (match the part number on the back of your panel) |
| **Time** | 25–35 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, safety glasses, Kapton tape |




1. Remove the 4 panel screws (sides of the panel).
2. Lay the panel face-down on the keyboard (cloth under it).
3. On the back: peel the tape over the connector, open the clip, pull the cable straight out.
4. Connect the new panel and TEST before screwing it down.

*Illustration: lid with bezel removed. Screw positions are typical. Count them on yours.*

**Cracked glass.** Wear safety glasses. Tape across a shattered panel before handling it.

Test first: if an external USB-C display works, the system board is fine and the fault is the panel or its cable.

## Procedure

1. ☐ Remove the bezel.
2. ☐ Remove the panel screws (count them; record the number in the notes box).
3. ☐ Tip the panel forward onto the keyboard (on a cloth).
4. ☐ Lift the tape over the display connector, open the latch and disconnect the cable.
5. ☐ Connect the new panel and test the picture **before** fixing it down.
6. ☐ Refit everything in reverse.

## Verify before closing the ticket

- Full image: no lines, dead pixels or uneven backlight.
- Brightness works across its whole range.
- Image steady through full lid travel.

# R13 Display (eDP) cable replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Flicker, lines or black screen that changes with lid angle; external display works |
| **Part** | Display (eDP) cable (look up by serial) |
| **Time** | 40–50 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, Kapton tape |

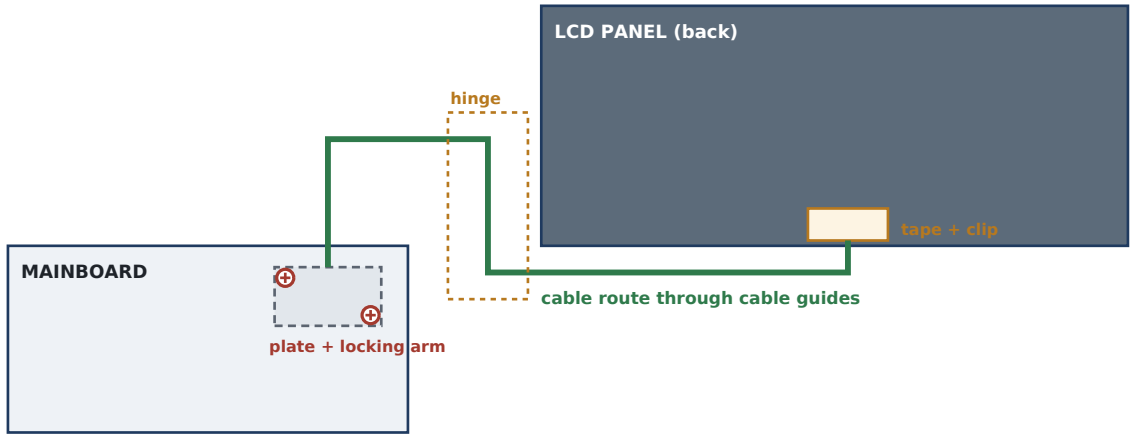



*Illustration: display cable path, mainboard to panel (route simplified). Photograph the real route before removing.*

## Procedure

1. ☐ Remove the bezel and panel; disconnect the display cable at the system board.
2. ☐ Free the cable from the back cover and hinge area; note every clip and tape point.
3. ☐ Route the new cable exactly along the old path and reconnect both ends.
4. ☐ Test at every lid angle before reassembly.

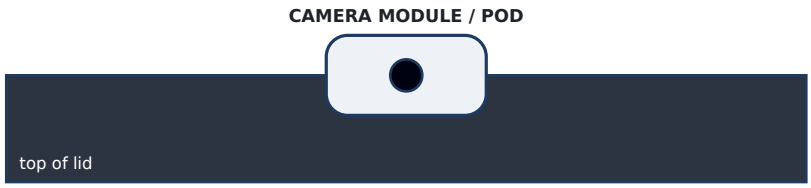
## Verify before closing the ticket

- Steady image at every angle: open and close the lid 20 times while watching.
- Camera still works if it shares the cable.

# R14 Camera module replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Camera not detected, black image, broken lens |
| **Part** | Camera module (look up by serial) |
| **Time** | 15–20 minutes |
| **Tools** | Plastic picks, plastic tweezers |




1. Bezel off first (it covers the camera's front).
2. Unplug the small camera cable with tweezers.
3. Lift the module by its board edges, never the lens.
4. Test in the Camera app before closing up.

*Illustration: camera at the top-center of the lid. Cap shapes simplified.*

Check first: privacy shutter open? Camera works after S1 and a powerwash (S3)?

## Procedure

1. ☐ Remove the bezel.
2. ☐ Lift any Mylar or tape over the camera module, disconnect its cable, and lift it off (it's usually held by adhesive).
3. ☐ Fit the new module, reconnect, and test before refitting.

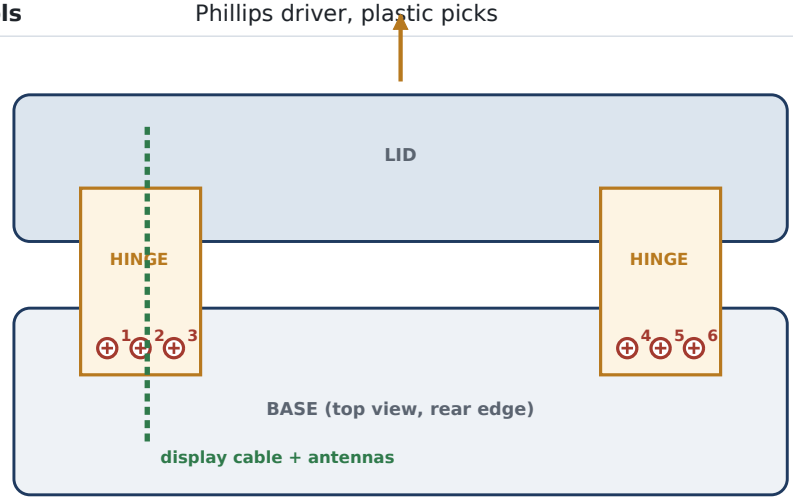
## Verify before closing the ticket

- Camera app shows a clear image.
- Privacy shutter fully covers the lens when closed.
- Works in a Google Meet test call.

# R15 Hinge replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Floppy lid, cracked hinge, grinding, lid won't stay up |
| **Part** | Hinges, left and right (look up by serial) |
| **Time** | 30–45 minutes |
| **Tools** | Phillips driver, plastic picks |




1. Battery out, display cable unplugged at the mainboard.
2. Unthread cable and Wi-Fi antennas from the guides FIRST.
3. Count the base hinge screws on first teardown.
4. Support the lid, then lift it away from the base.

*Illustration: hinge area seen from above with the D cover off. Positions approximate.*

Check first: tighten the hinge screws. A screw that spins without tightening means a stripped post: that's a cover replacement.

## Procedure

1. ☐ Separate the display assembly from the base (display assembly repair).
2. ☐ Remove the hinge-to-lid screws (count them; record the number in the notes box).
3. ☐ Lift the hinges off the lid.
4. ☐ Fit the new hinges (left and right differ) and reassemble.

## Verify before closing the ticket

- Lid holds any angle through its full travel.
- Both hinges feel equally stiff.
- Display steady and Wi-Fi strength normal (no pinched cables).

# R16 Keyboard / top cover replacement

LEVEL 3

| | |
|---|---|
| **Fixes** | Several dead keys, a dead row, liquid-damaged keyboard, cracked palm rest |
| **Part** | Upper case with keyboard (by keyboard language; look up by serial) |
| **Time** | 60–90 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, ESD strap, screw mat, Kapton tape |

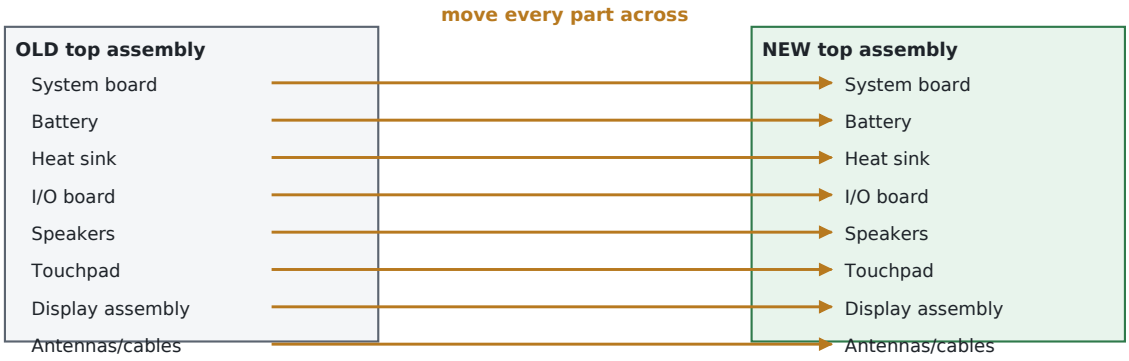



*Illustration: what moves to the new top assembly. Check what comes pre-fitted first.*

The keyboard is part of the upper case (palm rest). Everything else in the base moves to the new upper case: this is a full transplant.

## Procedure

1. ☐ Remove the bottom cover and the battery.
2. ☐ Remove every part listed below from the old upper case, in this order: battery, Wi-Fi card, I/O board, speakers, heat sink, system board, display assembly, touchpad.
3. ☐ Check what's pre-fitted on the new upper case, then transfer everything else.
4. ☐ Reconnect every cable, checking against your photos.
5. ☐ Reconnect the battery and test everything before closing up.

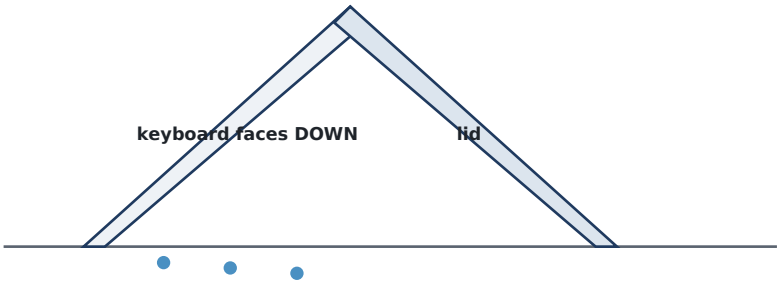
## Verify before closing the ticket

- Every key passes the S2 keyboard test.
- Touchpad, display, camera, ports, speakers and Wi-Fi all work.
- No screws left over.

# R17 Liquid spill response

LEVEL 3

| | |
|---|---|
| **Fixes** | Any liquid on or in the device |
| **Part** | Often none; sometimes keyboard/top cover or system board |
| **Time** | 10 min first response; 24–48 h drying |
| **Tools** | 90%+ isopropyl alcohol, soft brush, lint-free cloths |




1. Power off (hold 10 s), unplug.
2. Tent upside down so liquid drains OUT.
3. Open and disconnect the battery ASAP.
4. Dry 24–48 h. No hair dryer, no rice.

*Illustration: drain position after a spill.*

Spill-resistant keyboards resist drips, not a full cup. Speed matters more than anything.

**Do not power on to test.** Current through liquid causes the corrosion that kills boards.

## Procedure

1. ☐ Hold Power for 10 seconds to force off. Unplug.
2. ☐ Tent the device upside down (lid at 90°, keyboard down) to drain.
3. ☐ Open it and disconnect the battery as soon as possible.
4. ☐ Blot visible liquid; don't wipe it around.
5. ☐ Sugary or sticky spills: clean affected areas with a soft brush and isopropyl alcohol.
6. ☐ Dry open, battery disconnected, for 24 hours minimum (48 if soaked). No hair dryer, no rice.
7. ☐ Inspect for white or green corrosion; clean connectors.
8. ☐ Reconnect the battery, plug in, and run S2 fully.

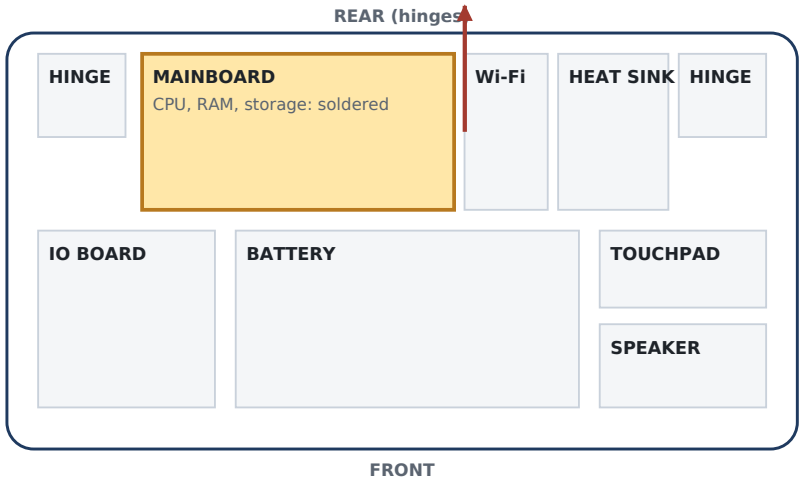
## Verify before closing the ticket

- All S2 tests pass.
- Sticky or dead keys: keyboard repair.
- Corrosion on the system board: Section 6.

# R18 System board replacement (lead tech)

LEVEL 3

| | |
|---|---|
| **Fixes** | Dead USB-C ports, no power with a good battery, failed storage, RAM errors, failed recovery |
| **Part** | System board (by configuration; look up by serial) |
| **Time** | 45–90 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, ESD strap |




1. Battery out first (and the Wi-Fi card if it's socketed).
2. Unplug: display, I/O board, touchpad, keyboard, speaker, antennas
3. Lift by the edges once every screw and cable is free.
4. Lift by the edges onto the ESD mat.

*Illustration: base interior, mainboard highlighted. Red arrow = lift side.*

**Identity and enrollment.** Storage, serial number and enrollment live on the system board. The lead tech updates the asset record, deprovisions the old device in the Admin console if needed, and plans re-enrollment. Ask the manufacturer whether the new board ships with the serial set.

Confirm first with S1, S2, S3 and a connector reseat. Most "dead boards" are a battery, cable or connector.

## Procedure

1. ☐ Remove the bottom cover, the battery, the Wi-Fi card, the heat sink.
2. ☐ Disconnect every cable from the board (photograph first).
3. ☐ Remove the system board screws (count them; record the number in the notes box).
4. ☐ Lift the board out by its edges.
5. ☐ Fit the new board, reconnect every cable, refit the parts removed, then run recovery (S3) if needed and enroll.

## Verify before closing the ticket

- Boots, enrolls and passes all of S2.
- Asset record and Admin console show the correct serial number.

# 6 Out of scope: escalate

These need soldering or board-level work, or parts that aren't sold. Tag the device with the ticket and send it to the lead tech.

| Fault | Why | Route |
|---|---|---|
| USB-C port loose or dead | Usually soldered to the system board | System board or warranty |
| Storage or RAM failure | Soldered (system-board level) | System board or warranty |
| Burn marks, melted plastic, burnt smell | Fire hazard | Lead tech immediately |
| Any repair needing a soldering iron or hot air | Program policy | CTL warranty or lead tech |

## CTL support

- **CTL support:** ctl.net/support (open a ticket with the serial number)
- **Parts:** CTL parts store / CTL support, by serial

# 7 Sources

Checked October 2026. The procedures are the guide author's adaptation of these sources for a student bench; they are not manufacturer-issued instructions.

- CTL product page and spec sheet for the Chromebook PX14E series (ctl.net) — specifications and variants.
- CTL support (ctl.net/support) — parts and service by serial.
- Standard Chromebook construction as documented in the other guides of this series.